

Time: 03 Hours

Marks: 80

- Note: 1. Question 1 is compulsory
 2. Answer any three out of the remaining five questions.
 3. Assume any suitable data wherever required and justify the same.

- Q1. (a) State examples of Deep Learning. 5
 (b) Explain optimizers. Why optimizers are required? 5
 (c) What are different loss functions and their use case? 5
 (d) What is the role of the encoder in an Autoencoder 5
- Q2. (a) Explain Classes of Deep Learning. 10
 (b) Explain working of gradient descent. 10
- Q3 (a) Discuss types of activation functions. 10
 (b) Explain Regularization in Autoencoders and types of Autoencoders. 10
- Q4 (a) Draw and explain the architecture of Convolutional Neural Networks. 10
 (b) Describe Pooling with suitable example. 10
- Q5 (a) Discuss Recurrent Neural Networks in detail. 10
 (b) Explain LSTM architecture. 10
- Q6 Write short note on 20
 (a) Application of Autoencoders
 (b) AlexNET: Architecture .
 (c) Explain the working of Gated Recurrent Unit
 (d) Generative Adversarial Network